

Milestone 1 report: a leak-free data pipeline for MILK10k

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1. Label strategy

Primary target, diagnosis 1. Lesions: Malignant 3,634 (69.4%), Benign 1,483 (28.3%), Indeterminate 123 (2.3%). All 123 Indeterminate lesions are actinic keratoses: the AK half of AKIEC, whose other 180 lesions (SCC in situ) are Malignant, so diagnosis_1 cannot be derived from an 11-class prediction (A1.1c). **Decision: keep Indeterminate as a third class.** An AK is a pre-cancer that is treated (cryotherapy, topical therapy): neither “reassure” (Benign) nor “excise” (Malignant). Merging it into Benign would teach the model to dismiss scaly red lesions, the look of SCC in situ; merging it into Malignant erases a distinct treatment path; dropping it hides the hardest cases from the test set while they still turn up in clinic. Its 87 training lesions get weight 14.05 in the loss and its recall is reported on its own.

Stretch target, 11 classes. DF 52, INF 50, VASC 47, BEN_OTH 44 and MAL_OTH 9 lesions. **Decision: keep all 11 and handle the rare ones with class weights**, not an “other” group: such a group would mix benign classes (BEN_OTH, DF, INF, VASC) with a malignant one (MAL_OTH), cutting the link to diagnosis_1, and DF and VASC look distinctive enough to deserve their own output. The price is large weights (MAL_OTH 66.7 against BCC 0.19), which make training noisier, and MAL_OTH cannot be evaluated reliably (A3.2), so its metrics are reported as not estimable. Mapping: outputs/label_map.json.

2. Split design and verification

Lesions are split first and images follow their lesion. split_lesions runs StratifiedGroupKFold (groups = lesion_id, strata = 11-class label refined by diagnosis_1, i.e. AKIEC split into its two values so both targets stay balanced) with 20 folds of 5%: 3 to test, 3 to validation, 14 to train. Seed 42, created 2026-10-05.

split	lesions	images	share
train	3,667	7,334	69.98%
val	786	1,572	15.00%
test	787	1,574	15.02%

Verified in code: 0 lesions shared by any pair of splits, every lesion’s two images in the same split, largest deviation of a class share from the overall share 0.21 pp (11 classes) and 0.11 pp (diagnosis_1). Over 10 seeds the sizes stay within 0.04 pp of 70/15/15 and the same seed gives the same split, but MAL_OTH reaches both val and test in only 7 of 10 seeds.

3. Imbalance on the train split

diagnosis_1: Malignant 2,542, Benign 1,038, Indeterminate 87 lesions, a ratio of **29.2 : 1**; 11 classes: BCC 1,765 against MAL_OTH 5, **353 : 1**. Both remedies are implemented: a class-weighted cross-entropy with weights $n / (k \cdot \text{count})$ from the train split (Benign 1.18, Indeterminate 14.05, Malignant 0.48; class_weights.json) and a WeightedRandomSampler. Over 20 train batches the sampler yields 197 / 218 / 225 images of Benign / Indeterminate / Malignant, against 187 / 10 / 443 with plain shuffling. A training run uses one of the two; both together would over-correct.

4. Data-quality issues and how they were handled

All 10,480 rows resolve to an image file, Image.verify() fails on none, and every image is 600x450 (B1). Every lesion has one dermoscopic and one clinical image, and age, sex, site and diagnosis agree between them (B4). Issues: (i) **anatom_site_general is missing for 37.3%** of images and the missing group looks like trunk lesions (NV share +11 pp), so it becomes an “unknown” category rather than an imputed site; age (0.4% missing) gets the train median; anatom_site_special (98% missing) is dropped. (ii) **Label leaks** never enter the model: diagnosis_2 to 4, diagnosis_confirm_type (72% of histopathology lesions are malignant against 2% otherwise), concomitant_biopsy and melanocytic. (iii) **Shortcut risk:** “altered” images are 4.5% of Benign, 17.5% of Indeterminate and 2.2% of Malignant images, so image_manipulation is excluded and test results will be split by it. File size carries no class signal (ROC-AUC 0.48 to 0.51, Part A) because every file shares one JPEG quality-75 setting. (iv) MAL_OTH has only 9 lesions in total.

5. Preprocessing and augmentation

Resolution 224x224. All images are 600x450, so 100% are larger than 224 and nothing is upsampled; 224 matches ImageNet-pretrained backbones, and 450 px leaves room for a 384 run later. eval_transform resizes the shorter side to 224 and centre-crops, which keeps the lesion’s proportions (a plain 224x224 resize would squash 4:3 into a square) at the cost of the outer 12.5% on the left and right; it is deterministic (asserted). **Normalization** from the 7,334 train images only: mean (0.686, 0.528, 0.477), std (0.123, 0.132, 0.152). **Views:** option (a), each image is a sample with its lesion’s label and the two predictions are averaged per lesion (aggregate_predictions); (b) one view only would throw away half the data and the second look a dermatologist uses; (c) both views per lesion is ready as LesionDataset for a two-branch model. Splits are by lesion, so per-image evaluation would not leak, but it would count every lesion twice, so metrics are per lesion. One train epoch loads in 40.9 s (batch 32, 2 workers).

augmentation (train only)	parameters	why the label does not change
RandomResizedCrop	224, area 0.8 to 1, aspect 3/4 to 4/3	framing varies; 80% of the frame stays, so the centred lesion is never cut out
Horizontal and vertical flips	p = 0.5 each	skin has no canonical left, right, up or down
RandomQuarterTurn	0, 90, 180, 270 degrees	orientation is arbitrary; lossless, no black corners
ColorJitter	brightness 0.1, contrast 0.1, no hue, no saturation	measured in A3.5: brightness 0.1 moves lesion brightness by 0.4x the Benign/Malignant gap; brightness 0.3 (the class example) moves it 1.2x and hue 0.05 shifts hue 1.3x the gap

6. Why accuracy on MILK10k is not real-world accuracy

MILK10k is a sample of the lesions that were biopsied, not of the lesions a dermatologist sees. 95.7% of its lesions were confirmed by histopathology, and a lesion is biopsied when a clinician suspects cancer, which is why 69% are malignant and 48% are BCC, while most lesions shown in a clinic are benign nevi and keratoses. The 224 lesions that entered without a biopsy confirm the mechanism: only 2% of them are malignant, because they were admitted for looking harmless. The benign class is unusual too: it holds the benign lesions that worried someone enough to remove them, not the easy ones. Two consequences follow for the final model. First, prevalence: accuracy, precision and negative predictive value all depend on the class mix. A constant “Malignant” answer already scores 69% accuracy here (A3.4), and in a clinic where cancer is rare the same sensitivity and specificity would turn most positive predictions into false alarms. Second, spectrum: the test measures how well the model separates lesions that were already suspicious enough to biopsy, not how it behaves in screening. So the final accuracy will only be read next to per-class recall (Malignant above all), specificity and the confusion matrix, which depend far less on the class mix, and any use outside this setting needs recalibration to the local prevalence and a test on consecutive, unselected lesions.

7. What could still go wrong

Residual leakage: MILK10k has no patient identifier, so two lesions of the same patient (same skin, camera and session) can sit in train and test; near-duplicate photographs across lesions were not searched for. **Bias:** one study team and set of devices, mostly

middle skin tones (dataset description), older patients (median 65 for malignant lesions) and biopsy-selected lesions. **Shortcuts:** “altered” photos, rulers, ink and gel bubbles must be audited with per-subgroup results. **What a clinician should know:** the model learned from biopsied lesions of one source; its probabilities need recalibration to the clinic’s prevalence; “Indeterminate” means actinic keratosis only; MAL_OTH is not evaluable; it has not been validated prospectively or on darker skin, so it is a second reader for triage and never a replacement for a biopsy.